Military Decision-Making Process (MDMP)

(HANDOUTS)

Signal Officer Advance Course (SOAC) Subject 3.2 • Signal Officer Basic Course (SOBC) Subject 2.2

Courses / Subjects	SOAC – Module 3 (Battalion/Brigade Operations Planning), Subject 3.2 Military Decision-Making Process SOBC – Module 2 (Battalion/Brigade Operations Planning), Subject 2.2 Military Decision-Making Process
Primary reference	PAM 3-0201, Commander and Staff Organization and Operations (promulgated 29 December 2023): Chapter IX, The Military Decision-Making Process, pp. 60–113
Supporting references	PAM 3-0201 Chapter II (Staff Duties and Responsibilities, pp. 13–24) and Chapter VII (Running Estimates, pp. 45–48)
How to use this handout	Read Parts I to IV before the session (pre-session work). Part V is your practical exercise scenario. Part VI holds the worksheets your group will fill in during the guided staff planning workshop and complete after the session.

Learning Objectives

Terminal Learning Objective (TLO). At the end of the session, the students will be able to apply the seven (7) steps of the Military Decision-Making Process (MDMP) by performing a guided staff planning activity in a simulated operational scenario, in accordance with applicable doctrine. (P3)

Enabling Learning Objectives (ELOs) and where this handout covers them:

SOAC	SOBC	Enabling Learning Objective	Covered in
3.2.1.1	2.2.1	Apply the MDMP by identifying staff roles and responsibilities during staff planning activities. (P3)	Part II; Part IV
3.2.1.2	2.2.2	Apply the seven (7) steps of the MDMP in sequence during a guided staff planning exercise. (P3)	Part III; Part V
3.2.1.3	2.2.3	Apply MDMP outputs to develop a basic staff planning product in a simulated operational scenario. (P3)	Part VI

POI content	Handout part
Core battalion staff roles and relationships (SOAC 3.2.1 / SOBC 2.2.1)	Part II
Seven (7) steps of the Military Decision-Making Process (SOAC 3.2.2 / SOBC 2.2.2)	Part I and Part III
Roles in MDMP support and staff planning (SOAC 3.2.3 / SOBC 2.2.3)	Part IV

Introduction

Every operation order a Signal officer supports, and every Signal annex a Signal officer writes, comes out of the Military Decision-Making Process. The MDMP is how a battalion or brigade commander and staff understand a mission, develop and test options, and turn the chosen option into an order. A Signal officer who knows the seven steps, the staff roles and the outputs of each step can put C4S requirements into the plan early, when they still shape the course of action, instead of after the plan is fixed. This handout follows PAM 3-0201 Chapter IX and is organized around the TLO and ELOs above.

Definition of Terms

Term	Meaning
Military Decision-Making Process (MDMP)	An iterative planning methodology to understand the situation and mission, develop a course of action, and produce an operation plan or order (PAM 3-0201, para 0902).
Course of action (COA)	A broad potential solution to an identified problem (para 0981).
Running estimate	A continuous evaluation of the current situation used to decide whether the current operation is going as expected and whether future operations are feasible. Each staff section keeps one; the S6's is the C4S (signal) running estimate (para 0702).
Warning order (WARNORD)	An order that gives subordinates advance notice so they can begin parallel planning and preparation. Issued after Steps 1, 2 and 6.
CCIR / PIR / FFIR	Commander's critical information requirement, made up of priority intelligence requirements (about the enemy and environment) and friendly force information requirements (about friendly forces) (paras 0948–0951).
EEFI	Essential element of friendly information: information to protect, not collect (para 0952).
IPB	Intelligence preparation of the battlefield: systematic evaluation of enemy, terrain, weather and civil considerations (para 0931).
Specified / implied / essential task	Tasks assigned by higher / unstated but required / must be done to accomplish the mission (paras 0934–0937).
War-gaming	A disciplined process, with rules and steps, that attempts to visualize the flow of the operation (para 09121).
Synchronization matrix	A record of war-game results showing a COA synchronized in time, space and purpose (para 09122).
Annex H (Signal)	The annex to the plan or order describing signal support, prepared by the S6 (para 0229e).
PACE plan	Signal planning term: primary, alternate, contingency and emergency means of communication for each requirement.
C4S	Command, control, communications and cyber systems.
AO / CP / GC / SOP	Area of operations / command post / grid coordinates / standard operating procedure.

Part I. Understanding the MDMP

1. What the MDMP is

The MDMP is one of the Army's three planning methodologies (with Army design methodology and troop leading procedures). PAM 3-0201 defines it as **an iterative planning methodology to understand the situation and mission, develop a course of action, and produce an operation plan or order** (para 0902). It helps leaders apply thoroughness, clarity, sound judgment, logic, and professional knowledge to understand situations, develop options, and reach decisions.

PAM 3-0201, paras 0901–0902

2. Characteristics

Characteristic	What it means for the staff	Para
Collaborative	Higher headquarters shares information through planning meetings and warning orders; units resolve conflicts with subordinate, adjacent, supporting and supported units before the order is published.	0903
Assessment starts in planning	The unit's assessment plan (measures of effectiveness and performance) is developed during the MDMP, not after the order is written.	0904
Drives preparation	Time analysis is done early. The commander may direct movement, task-organization changes and information collection through warning orders before the plan is complete.	0905
Works with Army design methodology	For unfamiliar or ill-defined problems, the commander may run Army design methodology before or alongside the MDMP. When time is short, the MDMP runs without it.	0906
Iterative	Staffs generally follow the seven steps in sequence but may revisit earlier steps as the situation develops.	0907

3. The seven steps at a glance

Each step has key inputs, a process and key outputs. The outputs of one step become the inputs of the next. Warning orders (WARNORDs) go out after Steps 1, 2 and 6 so subordinates can plan in parallel.

Step	Key inputs	Key outputs
1. Receipt of Mission	• Higher headquarters' plan or order, or a new mission anticipated by the commander	• Commander's initial guidance • Initial allocation of time → WARNORD 1
2. Mission Analysis	• Commander's initial guidance • Higher headquarters' plan or order • Higher headquarters' intelligence and knowledge products • Knowledge products from other organizations • Army design methodology products	• Problem statement • Mission statement • Initial commander's intent • Initial planning guidance • Initial CCIRs and EEFIs • Updated IPB and running estimates • Assumptions • Evaluation criteria for COAs → WARNORD 2

Step	Key inputs	Key outputs
3. COA Development	• Mission statement • Initial commander's intent, planning guidance, CCIRs and EEFIs • Updated IPB and running estimates • Assumptions • Evaluation criteria for COAs	• COA statements and sketches (tentative task organization, broad concept of operations) • Revised planning guidance • Updated assumptions
4. COA Analysis (War Game)	• Updated running estimates • Revised planning guidance • COA statements and sketches • Updated assumptions	• Refined COAs • Potential decision points • War-game results • Initial assessment measures • Updated assumptions
5. COA Comparison	• Updated running estimates • Refined COAs • Evaluation criteria • War-game results • Updated assumptions	• Evaluated COAs • Recommended COAs • Updated running estimates • Updated assumptions
6. COA Approval	• Updated running estimates • Evaluated COAs • Recommended COA • Updated assumptions	• Commander-approved COA and any modifications • Refined commander's intent, CCIRs and EEFIs • Updated assumptions → WARNORD 3
7. Orders Production, Dissemination, and Transition	• Commander-approved COA and any modifications • Refined commander's intent, CCIRs and EEFIs • Updated assumptions	• Approved operation plan or order • Subordinates understand the plan or order

Adapted from PAM 3-0201, Figure 9.1, p. 62

4. Full versus modified MDMP

The **full MDMP** is used when the commander has enough time and staff support to examine two or more COAs thoroughly, typically for an entirely new mission. In **time-constrained** conditions the commander decides how to abbreviate it; in extremely compressed situations the commander relies on more intuitive techniques such as the rapid decision-making and synchronization process. A unit must master the full MDMP before it can shorten it well (paras 0913–0914, 09192). See Part III, Step-by-step, and the time-saving techniques at the end of Part III.

Part II. Core Battalion Staff Roles and Relationships

1. What every staff officer does

The staff exists to **support the commander** (understand, visualize, describe, direct, lead and assess), **assist subordinate commanders, staffs and units**, and **inform units and organizations outside the headquarters** (paras 0202–0207). Common duties of every staff section include (para 0209):

- Advising and informing the commander
- Building and maintaining running estimates
- Making recommendations
- Preparing plans, orders and other staff writing
- Assessing operations
- Managing information within the area of expertise
- Identifying and analyzing problems
- Performing risk management
- Taking part in intelligence preparation of the battlefield (IPB)
- Exercising staff supervision over the area of expertise

2. Staff relationships

The staff works on the commander's behalf and derives its authority from the commander, who remains the decision-maker. Staff officers decide within their authority under established guidance and unit SOPs. The relationship depends on **open and honest communication**: staff officers give unbiased, truthful advice, and once the commander decides, they support and carry out the decision even when it differs from their recommendation (para 0220). Teamwork within and between sections is what produces coordinated operations (para 0221).

PAM 3-0201, paras 0220–0222

3. Battalion staff and their planning products

At battalion and brigade, the Executive Officer (Ex-O) is the commander's principal assistant and staff integrator; at division and higher, the Chief of Staff (CS) performs this role (para 0229c). The table summarizes the core staff, what each is responsible for, and the annexes each prepares.

Staff officer	Responsible for	Prepares / leads
Commander	Assigning tasks, delegating authority and making the final decisions. Everything the staff does or fails to do is the commander's responsibility.	Commander's intent, planning guidance, decisions
Executive Officer (Ex-O)	Principal assistant; manages the command post and headquarters; directs, coordinates and supervises the staff.	Staff planning timeline; quality control of all staff work
S1 – Personnel	Manning, personnel services and support, headquarters management; personnel aspect of sustainment.	Personnel portion of Annex F (Sustainment)
S2 – Intelligence	Intelligence on enemy, weather, terrain and civil considerations; chief of the intelligence cell.	Annex B (Intelligence); assists S3 with Annex L (Information Collection)
S3 – Operations	Operations, training and planning; chief of the movement and maneuver cell; integrates and synchronizes all warfighting functions; verifies plans and orders.	Annexes A (Task Organization), C (Operations), M (Assessment); publishes OPORDs, FRAGOs, WARNORDs

Staff officer	Responsible for	Prepares / leads
S4 – Logistics	Sustainment plans and operations: maintenance, supply, services, transportation.	Annex F (Sustainment)
S6 – C4S	Command and control, communications and cyber systems: information services, network transport, network operations and spectrum management; integrator for the mission command cell.	Annex H (Signal); with S3 and S2 input, Annex C and Appendix 12 (Cyberspace Electromagnetic Activities)
S7 – CMO	Civil, public and information support affairs; evaluates civil considerations during mission analysis and IPB.	Annex I (Civil-Military Operations)
S8 – Training	Training requirements of the command, focused on the unit's METL.	Training input to plans
Fire Support Officer (FSO)	Fires warfighting function; synchronizes fire support with the S3.	Annex D (Fires)
Engineer Officer / Chief of Protection	Protection warfighting function (CBRN, EOD, air defense, combat engineering, K-9, MP); principal advisor on force protection.	Annexes E (Protection) and G (Engineer)
Information Operations Officer	Synchronizes information-related capabilities; OPSEC and military deception input; leads the IO working group.	IO input to the synchronization matrix
Command Sergeant Major	Senior NCO; advises on enlisted matters; extends the commander's influence during critical events.	Advice and recommendations

Summarized from PAM 3-0201, paras 0229–0232

Part III. The Seven Steps of the MDMP

Step 1 – Receipt of Mission

Receipt of a mission, or its anticipation, triggers the MDMP. This step tells everyone a planning requirement is coming, determines how much time is available, and decides the planning approach (para 0916).

Process

1. **Alert the staff and other key participants.** Unit SOP prescribes who takes part in mission analysis.
2. **Gather the tools:** publications; the higher headquarters' OPLAN/OPORD, maps, terrain products and graphics; intelligence and assessment products; estimates of other agencies; SOPs; current running estimates; any Army design methodology products.
3. **Update running estimates.** Every staff section updates its running estimate as it gathers tools, and keeps doing so throughout the MDMP.
4. **Conduct initial assessment** of time and resources: time needed by the headquarters and subordinates, whether to abbreviate the MDMP, outside agencies to involve, and the staff's experience and level of rest or stress.
5. **Issue the commander's initial guidance:** time allocations, Army design methodology or straight to MDMP, how to abbreviate, liaison officers, authorized movement and information collection, collaborative planning times, initial information requirements, additional staff tasks.
6. **Issue the initial warning order:** at least the type of operation, general location, initial timeline, and any movement or information collection to start.

> **The one-third / two-thirds rule (para 0922)**
>
> As a rule, commanders allocate **at least two-thirds** of available time to subordinate units for their planning and preparation, and keep **one-third** for their own headquarters. Time, more than any other factor, determines how much detail the plan can have.
>
> Example: an order arrives 72 hours before execution. The battalion uses about 24 hours; companies get about 48 hours.
>
> The Ex-O turns the time allocation into a **staff planning timeline**: what product is due, who prepares it, who receives it, and the times and places of meetings and briefings (para 0923).

PAM 3-0201, paras 0916–0925, Figure 9.2

Step 2 – Mission Analysis

Mission analysis is **the most critical step of the MDMP**: if the problem is not understood well, no good plan can address it (para 0927). The commander and staff determine what the command must accomplish, when and where, and above all **why**.

Process (17 actions)

#	Action	Key points
1	Analyze the higher headquarters' plan or order	Higher commander's intent, mission, concept, assets, timeline; missions of adjacent, supporting and supported units; assigned area of operations. Clarify confusion at once; use requests for information.
2	Perform initial IPB	Evaluate enemy, terrain, weather and civil considerations. Products: threat COA models, modified combined obstacle overlay, high-value targets, event templates, initial PIRs.
3	Determine specified, implied and essential tasks	See the task box below.

#	Action	Key points
4	Review available assets and identify resource shortfalls	Task organization, command and support relationships, capabilities and limitations. Report shortfalls to higher headquarters.
5	Determine constraints	Restrictions placed by higher: a requirement to act or not to act (e.g., a no-fire area, a restrictive fire line).
6	Identify critical facts and develop assumptions	A fact is a statement of truth at a specified time. An assumption may be true but cannot be validated; it must be necessary for planning. Replace assumptions with facts as soon as possible.
7	Begin risk management	Identify and assess hazards now; develop controls during COA development.
8	Develop initial CCIRs and EEFIs	See the information requirements box below.
9	Develop the initial information collection plan	S3 develops the plan; S2 provides the management tools. Includes the communications plan for reports from collection assets to command posts (para 0956f).
10	Update plan for the use of available time	Compare own timeline with the higher and assumed enemy timelines; refine the staff planning timeline.
11	Develop initial themes and messages	S7/CMO officer reviews higher's themes and messages; commander approves.
12	Develop a proposed problem statement	Compare the current situation with the desired end state; list the issues that stand in the way.
13	Develop a proposed mission statement	Prepared by the Ex-O or S3. Answers who, what, when, where, why.
14	Present the mission analysis briefing	Each staff officer presents a summary of the running estimate. Outline in para 0970.
15	Develop and issue initial commander's intent	Purpose, key tasks and end state; normally three to five sentences.
16	Develop and issue initial planning guidance	Broad operational approach; which COAs to consider or not; guidance by warfighting function (Table 9.1).
17	Develop COA evaluation criteria, then issue a WARNORD	Criteria are set before COAs are compared to prevent bias; the Ex-O normally assigns weights. WARNORD includes the approved mission statement, intent, CCIRs and EEFIs, risk guidance and timeline.

PAM 3-0201, paras 0926–0980, Figure 9.3

Specified, implied and essential tasks (paras 0933–0937)

- **Specified task** – assigned to the unit by higher headquarters, usually in paragraphs 2 and 3 of the order, but also in paragraphs 4 and 5, annexes, overlays, or verbal guidance.
- **Implied task** – not stated but must be done to accomplish a specified task; found by analyzing the order, the enemy, terrain, civil considerations and doctrinal requirements.
- **Essential task** – a specified or implied task that must be executed to accomplish the mission. Essential tasks always appear in the mission statement.

Also identify **on-order** missions (to be executed at an unspecified time) and **be-prepared** missions (contingencies that may or may not be executed).

Information requirements (paras 0947–0953)

- **CCIR** (commander's critical information requirement) – information the commander identifies as critical to timely decision-making. Typically ten or fewer. Two kinds:
- **PIR** (priority intelligence requirement) – about the enemy and the operational environment; managed by the S2.

- **FFIR** (friendly force information requirement) – about the status of friendly forces and supporting capabilities; managed by the S3.
- **EEFI** (essential element of friendly information) – information to **protect**, not collect; if compromised it would jeopardize the mission. Not a CCIR, but approved by the commander and prioritized the same way.

Writing the mission statement (paras 0965–0967)
A short sentence or paragraph stating the essential task(s) and purpose: **WHO** (unit) – **WHAT** (essential task, as a tactical mission task) – **WHEN** (time or event) – **WHERE** (area, objective, GC) – **WHY** ("in order to …").
Example format: "11th Infantry Battalion (Exercise) **secures** San Isidro town proper **NLT** D-Day 0600H **in AO FALCON in order to** protect the population."

Step 3 – Course of Action Development

A COA is a broad potential solution to an identified problem (para 0981). The staff uses the mission analysis outputs to generate options that satisfy the commander's intent and planning guidance.

COA screening criteria (para 0983)

Criterion	Test
Feasible	Can accomplish the mission within available resources and their limitations.
Acceptable	Balances cost and risk with the advantage gained.
Suitable	Accomplishes the mission within the commander's intent and planning guidance.
Distinguishable	Differs significantly from the other COAs (scheme of maneuver, lines of effort, phasing, use of the reserve, task organization).
Complete	Includes the decisive operation, shaping and sustaining operations, offensive/defensive/stability tasks as required, and the conditions to be reached.

Process

1. **Assess relative combat power.** Compare friendly and enemy strengths and weaknesses for each element of combat power, starting with force ratios of maneuver units two levels down. It is largely subjective; include morale and training.
2. **Generate options.** Start with the decisive operation, then shaping and sustaining operations. Use the operational framework set by higher: deep–close–security, main and supporting effort, or decisive–shaping–sustaining.
3. **Array forces.** Determine the combat power each task needs, using historical minimum planning ratios as a starting point (below). Array generic units two levels down.
4. **Develop a broad concept:** purpose, where the commander accepts risk, critical events and phases, the reserve, information collection, scheme of fires, control measures, and the decisive operation with its task and purpose.
5. **Assign headquarters.** A headquarters normally controls two to five subordinate maneuver units.
6. **Develop COA statements and sketches.** The S3 prepares a statement and sketch for each COA covering who, what, when, where and why for each subordinate unit. The sketch includes boundaries, control measures, the main effort and the **locations of command posts and critical communications nodes** (para 09116).
7. **Conduct a COA briefing,** then the commander **selects or modifies COAs** for continued analysis.

Friendly mission	Position	Friendly : Enemy
Delay	—	1 : 6
Defend	Prepared or fortified	1 : 3
Defend	Hasty	1 : 2.5
Attack	Prepared or fortified	3 : 1
Attack	Hasty	2.5 : 1
Delay	Flank	1 : 1

Historical minimum planning ratios, PAM 3-0201, Table 9.2. For counterinsurgency, troop density of 20–25 security personnel per 1,000 residents is a common planning range (para 09104).

Step 4 – Course of Action Analysis (War Game)

War-gaming is **a disciplined process, with rules and steps, that attempts to visualize the flow of the operation** (para 09121). Each critical event is fought through the **action – reaction – counteraction** cycle. The results are refined COAs, a synchronization matrix, and a decision support template and matrix for each COA.

Process

1. Gather the tools (supervised by the Ex-O)
2. List all friendly forces
3. List assumptions
4. List known critical events and decision points
5. Select the war-gaming method
6. Select a technique to record and display results (synchronization matrix or sketch note)
7. War-game the operation and assess the results
8. Conduct a war-game briefing (optional)

Method	How it works	Best for
Belt	Divides the area of operations into belts across its width; analyzes events in each belt in sequence.	Terrain with well-defined cross-compartments, phased operations, enemy deployed in clear belts or echelons. The commander's usual preference.
Avenue-in-depth	Fights one avenue of approach at a time, starting with the decisive operation.	Offensive COAs, or defense in canalized terrain.
Box	Isolates one critical area (engagement area, crossing site, landing zone) and studies its critical events.	Time-constrained planning, hasty attacks, noncontiguous areas of operations.

PAM 3-0201, paras 09130–09137. In stability tasks, each method can be applied to lines of effort instead of ground.

War-gaming rules (para 09152)

- Stay objective; ignore personalities and "what the commander wants."
- Do not defend a COA just because you developed it.
- Record advantages and disadvantages as they come up.
- Keep checking feasibility, acceptability and suitability; reject a COA that fails.
- Do not draw conclusions without supporting facts.

- **Do not compare COAs during the war game.** That is done in Step 5.

Who does what in the war game: the **Ex-O** is the unbiased controller and ensures at least the decisive operation is war-gamed; the **S3** plays the friendly maneuver commander; the **S2** plays the enemy commander and other threat groups; the **S6** assesses network operations, spectrum management, network defense and information protection feasibility for each COA; the **FSO**, **chief of protection**, **S1** and **S4** test their functions; **recorders** capture results on worksheets (paras 09153–09170).

Step 5 – Course of Action Comparison

COA comparison objectively weighs COAs against the criteria the commander and staff agreed on, to find the COA with the best likelihood of success (para 09171).

Process

1. **Conduct advantages and disadvantages analysis** – each staff officer judges each COA from his or her functional area, using the evaluation criteria.
2. **Compare COAs** – often with a decision matrix.
3. **Conduct a COA decision briefing** – the staff recommends a COA. If the staff cannot agree, the Ex-O decides which COA to recommend.

Using the decision matrix (paras 09173–09175)

Rank each COA against each criterion (1 = best). The Ex-O assigns each criterion a weight based on its relative importance. Multiply each rank by the weight and add across. **The lowest total is the most desirable COA.** The matrix is a tool, not the decision: it rests on subjective judgments, and comparing COAs criterion by criterion is often more useful than the totals.

Criteria →	Protection of population	Continuity of C2	Flexibility	Simplicity	Total
Weight	2	2	1	1	
COA 1	1 (2)	1 (2)	2 (2)	1 (1)	5 (7)
COA 2	2 (4)	2 (4)	1 (1)	2 (2)	7 (11)

Worked example: unweighted rank, with the weighted value in parentheses. COA 1 has the lower total in both, so it is preferred, but COA 2 is more flexible, which the staff should tell the commander.

The selected COA should also pose the least risk, put the unit in the best position for future operations, give subordinates the most room for initiative, offer the most flexibility, and provide the safest environment for civilians (para 09176).

Step 6 – Course of Action Approval

After the decision briefing, the **commander approves a COA**, modifies one, or rejects them all (in which case COA development starts again). A modified or new COA is war-gamed before the staff recommends it (para 09179). The commander then issues **final planning guidance**, including refined intent, additional CCIRs, priorities, the risk the commander will accept, and guidance on orders and rehearsals. The staff sends a **WARNORD** with the area of operations, mission, intent, updated CCIRs and EEFIs, concept, tasks to subordinates and the final timeline (paras 09180–09182).

Step 7 – Orders Production, Dissemination, and Transition

The staff turns the approved COA into the plan: **the COA statement becomes the concept of operations, and the COA sketch becomes the basis of the operation overlay** (para 09183). The S3 drafts the order with the Ex-O directing the staff principals.

- **Plans and orders reconciliation** (internal): checks that the base order and all annexes agree, especially intent, mission and CCIRs against the concept and supporting schemes.
- **Plans and orders crosswalk** (external): compares the order with higher and adjacent orders for unity of effort.
- **Approval:** the commander approves the plan or order. Annexes are reviewed before the base order is signed.
- **Transition:** responsibility for the plan passes from the plans cell to the current operations cell; subordinates acknowledge receipt and give **confirmation briefings**.

PAM 3-0201, paras 09183–09190, Figure 9.15

Planning in a time-constrained environment

The commander decides how to modify the MDMP. Time is the one non-renewable resource. Up-to-date **running estimates** let the staff move straight into COA development (paras 09195–09199). Time-saving techniques (paras 09200–09205):

- Increase the commander's involvement so decisions are made without waiting for formal briefings.
- Limit the number of COAs; in extreme cases, develop only one COA against the most likely enemy COA.
- Maximize parallel planning: a verbal WARNORD now beats a written order an hour from now.
- Increase collaborative planning with higher and subordinate headquarters.
- Use liaison officers who have the commander's full trust.

Part IV. Roles in MDMP Support and Staff Planning

The commander has the most important role: the commander guides the planning effort, takes part during critical periods, and decides (para 0909). The Ex-O manages and coordinates the staff's work and provides quality control (para 0911). The staff helps the commander understand the situation, make decisions, and synchronize them into a complete plan or order (para 0912). The matrix shows the lead roles at each step.

Step	Commander	Ex-O	S2	S3	S6 (Signal)
1	Initial assessment; initial guidance	Staff planning timeline	Gathers intelligence products; updates estimate	Alerts staff; issues WARNORD 1	Updates C4S running estimate; checks network status and spectrum
2	Approves mission statement, CCIRs, EEFIs; issues intent and planning guidance	Proposes mission statement; sets COA evaluation criteria and weights	Initial IPB; threat COA models; PIRs	Specified/implied tasks; constraints; information collection plan; FFIRs	C4S facts, assumptions, shortfalls; recommends EEFIs; communications plan for collection assets
3	Guides and limits COAs; selects COAs for analysis	Supervises development	Updated IPB; most likely and most dangerous threat COAs	COA statements and sketches	Locates CPs and critical communications nodes on each sketch; checks supportability
4	Oversees; may modify COAs	Unbiased controller of the war game	Plays the enemy; refines PIRs and event template	Plays friendly maneuver commander; recorder	Assesses network operations, spectrum management, network defense and information protection for each COA
5	Receives decision briefing	Decides recommendation if staff disagrees	Compares COAs in intelligence terms	Leads comparison	Compares COAs for C4S supportability; ranks them in the S6 estimate
6	Approves COA; final planning guidance	Directs final WARNORD	—	Issues WARNORD 3	Updates C4S priorities and timeline
7	Approves and signs the order	Directs orders production; reconciliation and crosswalk	Annex B; part of Annex L	Drafts base order, Annexes A, C, M	Annex H (Signal); input to Annex C and Appendix 12

Compiled from PAM 3-0201 Chapter IX and paras 0229–0231. Other staff (S1, S4, S7, FSO, chief of protection) contribute in their functional areas at every step.

The Signal officer in the MDMP

Whether serving as the battalion S6 or supporting the S6 as a Signal platoon leader, the Signal officer's contribution to the MDMP comes through the **C4S running estimate**. PAM 3-0201 calls for the S6 to:

- Evaluate **communication system needs against available assets** for each war-gamed COA, detect potential flaws, and offer ways to eliminate or reduce them (para 09155).
- Assess the feasibility of **network operations, spectrum management operations, network defense and information protection** for each COA (para 09155).
- Contribute to electronic warfare products: electronic warfare target list, electronic attack tasking and requests, and the electronic warfare and attack guidance matrices (para 09159).
- Plan the **communications for transmitting reports** from collection assets to command posts in the information collection plan (para 0956f).

- Ensure **command posts and critical communications nodes** appear on COA sketches (para 09116) and that **information systems node movement times** are determined in the war game (Table 9.4).
- Plan **primary, alternate, contingency and emergency (PACE)** means for each COA, and show how each COA changes them.
- Recommend **EEFIs** that protect friendly communications, such as frequencies, call signs, net structure and relay locations.
- Prepare **Annex H (Signal)** and, with the S3 and S2, contribute to Annex C and Appendix 12 (para 0229e).

Running estimate format (PAM 3-0201, Figure 7.1)

Paragraph	Content (applied to C4S)
1. Situation and considerations	Area of interest; terrain, weather, enemy, friendly C4S resources and civil considerations as they affect communications; facts and assumptions.
2. Mission	The restated mission from mission analysis.
3. Courses of action	Friendly COAs war-gamed; enemy actions that affect C4S; evaluation criteria (all staff use the same criteria).
4. Analysis	Each COA analyzed against the criteria; C4S issues, risks and deficiencies.
5. Comparison	COAs ranked for each key C4S consideration, using a decision matrix.
6. Recommendations and conclusions	Most supportable COA from the C4S view; prioritized issues and mitigation.

Self-Check Questions

1. List the seven steps of the MDMP in order.
2. What are the two key outputs of Step 1, and what order is issued at the end of it?
3. Why is mission analysis considered the most critical step?
4. Explain the one-third / two-thirds rule.
5. Distinguish specified, implied and essential tasks. Which appear in the mission statement?
6. What are the five COA screening criteria?
7. In the war game, who serves as the unbiased controller, and who plays the enemy commander?
1. What does the S6 assess for each war-gamed COA?
2. Name the three war-gaming methods and when each is best used.
3. In a decision matrix, does the lowest or highest total indicate the most desirable COA?

Summary

The MDMP turns a mission from higher headquarters into an approved plan or order through seven steps: receipt of mission, mission analysis, COA development, COA analysis (war game), COA comparison, COA approval, and orders production, dissemination and transition. The commander drives the process; the Ex-O manages it; each staff officer contributes through a running estimate. Mission analysis is the most critical step, and the output of each step feeds the next. For the Signal officer, the MDMP is where C4S requirements, shortfalls, EEFIs and the Signal annex are built into the plan, so that the commander's plan can actually be commanded and controlled.

References

- PAM 3-0201, Commander and Staff Organization and Operations (29 December 2023): Chapter II, Staff Duties and Responsibilities; Chapter VII, Running Estimates; Chapter IX, The Military Decision-Making Process.
- Program of Instruction, Signal Officer Advance Course, Subject 3.2, and Signal Officer Basic Course, Subject 2.2.